



Practical: 5 Linux based

import OS

import subprocess

```
def run-command (command):
```

```
    subprocess.run (command, shell=True,  
                    check=True)
```

```
# install Java
```

```
run-command('sudo apt-get update')
```

```
run-command('sudo apt-get install -y default-jdk')
```

```
# Download & install Hadoop
```

```
run-command('wget http://downloads.  
apache.org/hadoop/common/hadoop-3.3.4')
```

```
run-command('tar -xzf hadoop-3.3.4.tar.gz')
```

```
run-command('sudo mv hadoop-3.3.4 /usr/  
local/hadoop')
```

```
# set Hadoop environment
```

```
OS.environ['HADOOP_HOME'] = '/usr/local/hadoop'
```

```
OS.environ['PATH'] += ': /usr/local/hadoop/bin:  
/usr/local/hadoop/sbin'
```



#set HIVE environment

```
OS -enviroment[HIVE_HOME]='(usr)/local/(hive)'  
OS -enviroment[PATH]='':(usr)/local/hive/bin
```

start Hadoop service

```
run_command('start-dfs.sh')  
run_command('start-yarn.sh')
```

Hive schema

```
run_command('schematool -dbType derby  
-initSchema')
```

start HIVE metastore, hive server2

```
run_command('nohup hive --service metastore  
run_command('nohup hive --service hive server2
```




Python

```
from pyhive import hive
import pandas as pd
```

```
conn = hive.Connection(host='localhost', port=10000,
                        username='hadoop')
cursor = conn.cursor()
```

```
cursor.execute("CREATE DATABASE IF NOT EXISTS test-db")
```

```
cursor.execute("USE test-db")
```

```
cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS employees
(id INT,
 name STRING,
 age INT,
 department STRING,
)
```

```
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
```

```
""")
```

```
cursor.execute("INSERT INTO employees values (1, 'John Doe', 30, 'HR')")
```

```
cursor.execute("INSERT INTO employees values (2, 'June Doe', 25, 'Finance')")
```



```
cursor.execute("CREATE INDEX IF NOT EXISTS  
idx_name ON TABLE empless (name)  
AS 'compact' WITH DEFERRED  
REBUILD")
```

```
cursor.execute("Drop INDEX IF EXISTS idx_name  
ON empless")
```

```
cursor.execute("Drop VIEW IF EXISTS employee")  
cursor.execute("DROP DATABASE IF EXISTS  
test_db")
```

```
cursor.close()  
conn.close()
```